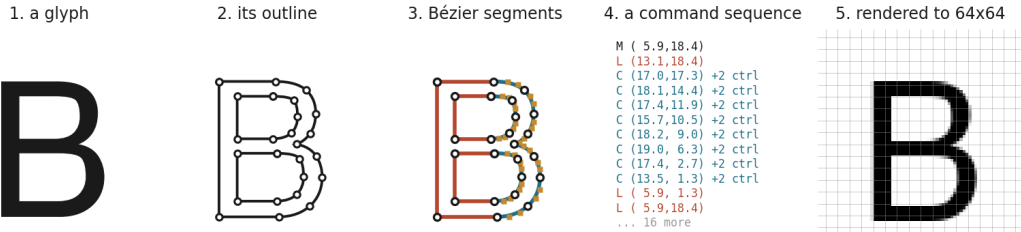


DeepVecFont-v2: Reproduction and Architectural Sweep

Ben Sapirstein 205987126, Dvir Yom Tov 209399872

Final project, part 2. Generative Models for Text and Images, Reichman University.

Paper: Wang, Wang, Yu, Zhu, Lian. *DeepVecFont-v2: Exploiting Transformers to Synthesize Vector Fonts with Higher Quality*. CVPR 2023.



The letter B, from a font in the English test split, at five stages of the pipeline: the glyph as a designer drew it, its outline, the curves and lines that outline is built from, the command sequence the model reads and writes, and the 64×64 rendering the score is computed on. The model works in panel 4. The number everyone reports is measured in panel 5.

Abstract

A vector font is a set of drawing programs rather than a set of pictures: one command sequence per character, each of which has to stay editable and legible at any size. That places font synthesis closer to program synthesis than to image synthesis, and it is why the models that do it well read and write both modalities of a glyph at once, the rendered image and the outline. DeepVecFont-v2 (Wang et al., CVPR 2023) is the current form of that idea: a Transformer encoder-decoder that sees a handful of reference glyphs, writes the command sequences for the rest of the alphabet, and refines its own draft in a second decoding pass.

We reproduce the model on Chinese and English, check the reproduction against the authors’ released weights, and run 21 single-factor changes to the architecture against it.

The measurement came first. Retraining the unmodified baseline three times under different random seeds moves the score by 0.0097 on Chinese, so that is the margin of error any candidate has to beat.

The change that worked is E9. The released code adds a full-strength Gaussian perturbation to the encoder’s output on every training step; E9 halves it, from $\sigma = 1.0$ to $\sigma = 0.5$, and leaves the model otherwise untouched. It improves Chinese reconstruction on all three metrics and at every seed, by 0.0040 in Error and 0.0271 in s-IoU. Both gains are smaller than the margin of error, so we report E9 as a consistent direction rather than as a gain of a stated size. Of the other twenty changes, one degrades and the rest are indistinguishable from noise.

Introduction

Automatic font generation is an old goal with a recent shape. A designer draws a few characters and would like the rest of the alphabet to follow in the same style, and for scripts with large character sets the argument is stronger: a Chinese font runs to tens of thousands of glyphs, all of which have to agree with each other. Deep generative models handle the raster version of this task well. The vector version, where the output has to be an outline a designer can open and edit, has been harder, because that output is a variable-length sequence of discrete drawing commands rather than a fixed grid of pixels.

DeepVecFont (Wang and Lian, ACM TOG 2021) set up the approach this project builds on. It encodes both modalities of a glyph on the argument that they carry complementary information: the image gives a global reading of shape, the command sequence gives the

precise local structure a designer actually edits. It generates the sequence with an RNN and a mixture density head, then corrects the raw output by rendering it through a differentiable rasterizer and pulling the result toward the synthesized image. That correction is what makes the outlines usable, and it is also the weak point, because it fits the noise in the image guiding it. Its published outlines show corners that are over-smoothed in some places and joins that are under-smoothed in others.

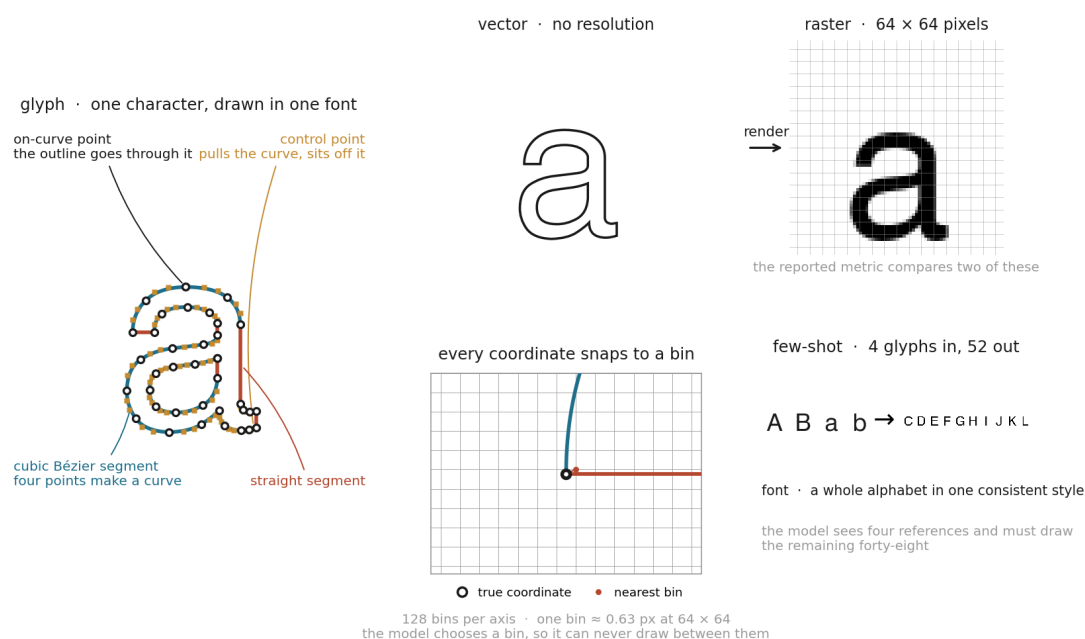
DeepVecFont-v2 (Wang, Wang, Yu, Zhu, Lian, CVPR 2023) keeps the dual-modality premise and replaces the machinery around it. Transformers take over from RNNs, which forces a change to the outline representation itself: the ordinary SVG encoding lets each command inherit its starting point from the one before, a convention that suits a recurrent model reading the sequence in order and starves a Transformer reading it in parallel. So v2 writes every command with its own start and end, and adds a relaxation representation so gradients can pass through the discrete arguments. A Bézier alignment loss samples auxiliary points along the predicted curve and pulls them onto the target curve, giving the model a signal about curve shape that control-point supervision alone does not carry. And the post-hoc rasterizer correction gives way to a second Transformer decoder that refines the whole sequence in context.

The gain from that redesign is 0.006 in reconstruction error on Chinese and 0.004 on English (section 2). It is worth knowing that number before reading any of ours. Where a year of work on a problem moves a metric bounded by 1 through six thousandths of a point, the question worth asking about any further change is whether the difference it makes can be told apart from retraining the same model with a different random seed. Section 4 takes that question literally and answers it by measurement, and the answer governs how every result in section 5 is read.

This report covers the architecture (section 1), the paper’s own results (section 2), our reproduction of them (section 3), the changes we tested (section 4), what those changes returned (section 5), and what we take from it (section 6).

1. Original architecture

DeepVecFont-v2 generates a full vector font from a few reference glyphs. It uses two modalities at once: a 64×64 raster image of each glyph and the glyph’s outline as a sequence of drawing commands.



*A **glyph** is one character drawn in one style; a **font** is a whole alphabet drawn consistently.*

Its **outline** is built from segments, either straight or **cubic Bézier**, where each curve is fixed by two **on-curve points** it passes through and two **control points** it does not. To **render** is to turn that outline into pixels. The model never emits a raw coordinate: it picks one of 128 **bins** per axis, so it can only place a point on that grid. And it works **few-shot**, seeing four reference glyphs and having to draw the other forty-eight.

Encoders. A CNN encodes the reference images. A Transformer encodes the reference command sequences. The two are fused into one latent vector per font.

Decoders. Three heads read that latent. A CNN decodes a raster image. An autoregressive Transformer decoder emits the command sequence one token at a time. A second Transformer decoder then refines the whole sequence in parallel. **The refined sequence is what gets scored**, which matters later.

Representation. Each command is a type plus 8 coordinate arguments. Coordinates are quantized to a fixed grid of bins and predicted as a classification over bins, which is what makes the model autoregressive rather than a regressor. The paper adds a relaxation representation so gradients can flow through the discrete arguments, and a Bézier alignment loss that pulls sampled points on the predicted curve toward the target curve.

Loss. A weighted sum of an image L1 term, a command-type cross-entropy, a coordinate-bin cross-entropy, the Bézier alignment term, and a KL term on the latent.

Key hyperparameters, as shipped and as used by us:

	Value
Optimizer, learning rate	Adam, 2e-4
Batch size	32
Image size	64×64
Coordinate bins	128
Transformer width, encoder depth	512, 6
Reference glyphs	4 (English), 8 (Chinese)
Max sequence length	51 (English), 71 (Chinese)
Loss weights	image L1 10, Bézier 0.01, point 0.01
Parameters	136,055,207

1.1 Where the released code differs from the paper

We read the paper against the code. Four differences matter, and three of them became experiments in section 4.

Paper says	Code does
Coordinate arguments in 256 dimensions	quantizes to 128 bins
Self-refinement is a 2-layer decoder	builds 1 layer
Bézier loss weighted at 1.0	loss_w_aux = 0.01, 100× lower
The $N(0, I)$ perturbation is an inference-time device	applied in training, validation and test alike

Two more, found late, affect how our numbers compare to the published ones. The paper uses 10 samples per glyph on English and 50 on Chinese; we used 50 on both, which flatters our English figures. And the paper enlarges the Chinese training set ten times by affine augmentation, while our built dataset holds 1,272 entries over 212 base fonts, which is six times.

One more thing we found is not a paper discrepancy but is worth recording. The encoder builds cross-attention modules and never calls them. They amount to 23,204,402 parameters, 17% of the model, trained by the optimizer and used by nothing.

2. Paper results

The paper reports **reconstruction error**, written “Error” in its tables. The predicted outline is rendered to a 64×64 image, the ground truth glyph is rendered the same size, and the metric is the mean absolute difference between the two images. It is measured per glyph and averaged. Lower is better. The model draws several candidate outlines per glyph and the one with the highest overlap against the target is kept, so the reported number is a best-of-N score rather than a single draw.

Model	Error, English	Error, Chinese
DeepSVG	0.125	0.167
DeepVecFont	0.056	0.086
DeepVecFont-v2	0.052	0.080

Note the scale. The paper’s whole improvement over its own predecessor on Chinese is 0.006. On a 64×64 image that is about 25 of 4,096 pixels. Every change we test in section 4 produces a difference around that size or smaller, and section 5 is largely about whether such differences can be measured at all.

The paper also reports two ablations. Its English ablation spans 0.0069 in total across four rows. Its sampling-point study spans 0.0037 across six settings.

The metric from the course material. We were asked to add one metric studied in class alongside the paper’s original. We use **SSIM**, which compares local structure rather than per-pixel difference and is standard for image quality. We implemented it in `eval_reconstruction_error.py` and checked it against `skimage` to $1e-16$. We also report **s-IoU**, the intersection over union of the two rendered shapes, because the vector font literature uses it and because it turned out to disagree with L1 often enough to be worth carrying.

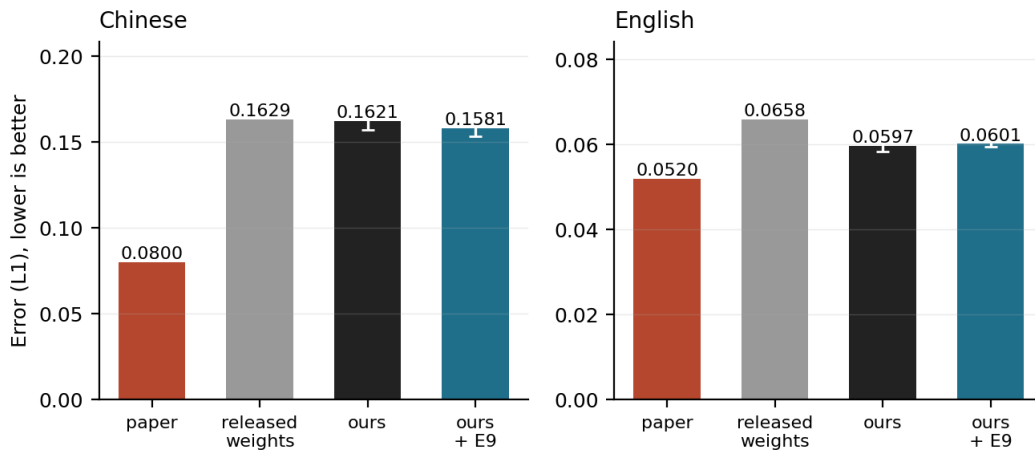
3. Reconstruction results

All numbers below are over the same 34 test fonts, best-of-50 decoding, with the ground truth taken from the dataset’s pre-rendered images.

	Error (L1)	s-IoU	SSIM
Paper, Chinese	0.080	not reported	not reported
Released weights, Chinese, epoch 600	0.1629	0.3225	0.4373
Our reconstruction, Chinese , 3-seed mean, 150 epochs	0.1621	0.2681	0.4425
Paper, English	0.052	not reported	not reported
Released weights, English, epoch 600	0.0658	0.7029	0.7181
Our reconstruction, English , 3-seed mean, 630 epochs	0.0597	0.7309	0.7374

Error is lower-is-better; s-IoU and SSIM are higher-is-better.

Our reconstruction matches the released weights; neither reaches the published number



We did not reach the published numbers. We did reproduce the released model, and those are different claims.

On Chinese our reconstruction sits 0.0008 from the authors' own checkpoint, on a quarter of the training budget. On English ours scores better than all three released checkpoints.

Training longer does not close the remaining distance. The released checkpoint has four times our budget and lands in the same place, and our own Chinese runs taken to 600 epochs select their best checkpoints at epochs 150 to 200 and score 0.1583.

The distance to 0.080 is therefore a property of the evaluation rather than of our training. The comparison the brief asks for is the one against the released checkpoint, and that comparison is the table above.

A 0.16 Chinese Error is an average, and an average hides what the tail looks like. Section 5.4 shows both tails from the same six freshly-decoded test fonts, picked by L1 rather than by eye, alongside the qualitative baseline-versus-E9 comparison they belong next to.

Every English figure here used 50 samples per glyph where the paper uses 10. Scoring one fixed checkpoint both ways puts that difference at 0.0027, so our English column is optimistic by roughly that much.

4. Improved architecture

4.1 Measuring the margin of error

The paper's own margin is 0.006. Before trying to beat it we checked whether we could measure anything that small. We trained the unmodified baseline three times, changing only the random seed, and looked at how far apart the three scores landed.

They span 0.0097 on Chinese and 0.0038 on English, and 0.0315 on Chinese s-IoU. That is the margin of error. A change that moves the score by less than this cannot be told apart from a lucky seed.

Two rules were written down before any candidate ran. A candidate has to beat the margin of error to count as a gain, and it has to move the same way at all three seeds, so one good run is not a result. Both were fixed before any number arrived; where we later overruled one, we recorded the overrule and the reason.

4.2 Candidate changes

We ran 21 distinct single-factor changes. Each moves one thing and holds everything else at the baseline value. Every new option defaults to the released behaviour, and a preflight

script asserts that, so a candidate’s default reproduces the baseline exactly.

Three candidates come from the paper-versus-code differences in section 1.1, and those are the ones we expected most from, because each asks whether the paper’s stated design or the shipped code is better.

Category from the brief	What we changed
Add normalization layers	terminal LayerNorm on the encoder (E1); batch norm in the image branch (E2)
Change the encoder or decoder	encoder width (E4); encoder depth 6→8 (E16); decoder feed-forward width 1024→2048 (E17)
Change the latent dimension	bottleneck 512→256 and →1024 (E5)
Add residual or attention layers	refinement decoder depth 1→2 and →3 (E3); encoder depth (E16)
Modify the loss function	Bézier weight 0.01→0.1/0.3/1.0 (E7); ordinal label smoothing (E8); KL weight (E12)
Change the noise schedule	encoder noise σ at train time (E9); warmup-cosine learning rate (E14)
Add regularization	dropout (E10); AdamW (E11); weight EMA (E15)

All seven categories in the brief are covered, and one representative of each was rerun at three seeds. That last part is the point: a coverage table built from single runs would show breadth and prove nothing, for the reason section 6.3 gives.

Two candidates did not survive contact with the code, and we report them rather than dropping them. Instance normalization cannot compute a per-instance variance at the model’s [32, 1024, 1, 1] bottleneck, so E2 ran with batch norm only. And we never ablated the 17% dead parameters, because removing them shifts the random stream for every module built afterwards, which would make it partly an initialization change.

4.3 The change that worked: encoder noise at training time (E9)

Between the encoder and the decoders, the released code adds a full-strength Gaussian perturbation to the font’s latent vector: `x + torch.randn_like(x)`, on every forward pass.

The paper introduces that perturbation as an inference-time device. Sampling around the latent is how the model offers a designer several drafts of the same glyph instead of one, which the paper describes as simulating the uncertainty of human design. The code applies it during training as well, where it acts as a regularizer nobody asked for: the encoder is made to reconstruct a font from a latent pushed a full unit away from the one it just computed.

E9 halves that noise during training, $\sigma = 1.0$ down to $\sigma = 0.5$, and keeps it at full strength at test time. The model still gets the sampling variety the paper wants, and it now trains against a latent close to the one it actually produced. Nothing else moves: same architecture, same parameter count, same loss, same schedule. That makes it a change to training dynamics rather than to the architecture, and we report it as one.

One detail matters for reading the rest of the sweep. The noise is still drawn even when scaled down, so the random number stream stays aligned across every value of σ . Otherwise changing σ would also change the initialization, and the comparison would be measuring two things at once.

5. Improved results

5.1 Comparison with the paper and the released model

	Paper	Released	Our reconstruction	E9, $\sigma=0.5^\dagger$
Chinese , Error (L1)	0.080	0.1629	0.1621 ± 0.0047	0.1581 ± 0.0046
Chinese , s-IoU	—	0.3225	0.2681 ± 0.0122	0.2952 ± 0.0191
Chinese , SSIM	—	0.4373	0.4425 ± 0.0057	0.4479 ± 0.0094
English , Error (L1)	0.052	0.0658	0.0597 ± 0.0021	0.0601 ± 0.0007
English , s-IoU	—	0.7029	0.7309 ± 0.0065	0.7305 ± 0.0041
English , SSIM	—	0.7181	0.7374 ± 0.0076	0.7358 ± 0.0005

Our columns are three-seed means $\pm$ sample standard deviation. † E9 improves Chinese on all three metrics with the same sign at every seed (section 5.2). The effect is smaller than the margin of error we committed to in advance, so we report it as a direction rather than as a gain of a stated size, and section 5.3 shows it does not carry over to English.

5.2 E9 on Chinese

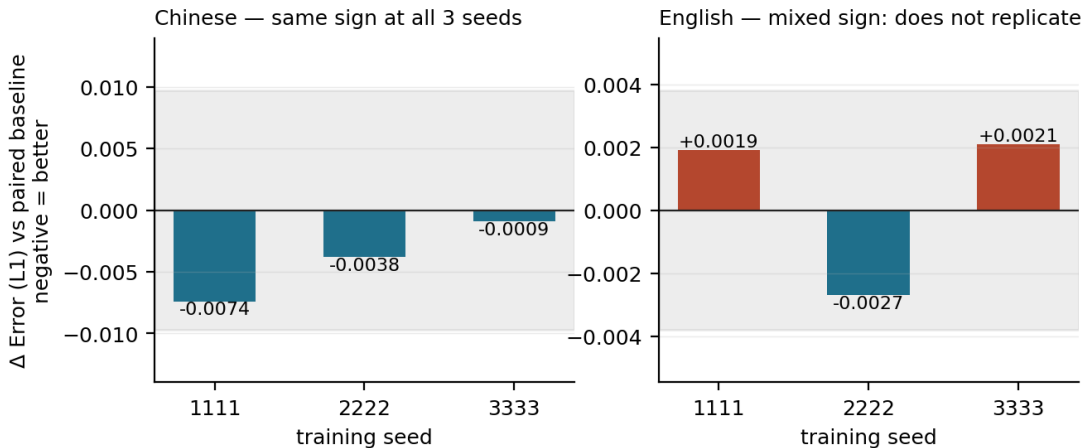
Chinese, best-of-50, each candidate seed paired against the baseline seed of the same number.

Seed	E9 Error	Δ vs paired baseline	E9 s-IoU	Δ
1111	0.1588	-0.0074	0.2870	+0.0325
2222	0.1531	-0.0038	0.3170	+0.0454
3333	0.1623	-0.0009	0.2816	+0.0035
Mean	0.1581	-0.0040	0.2952	+0.0271

E9 improves both metrics at every seed, six agreeing signs out of six, on two metrics that barely correlate with each other ($r = -0.335$). Neither mean beats the margin of error, though: 0.0040 against 0.0097 on Error, 0.0271 against 0.0315 on s-IoU.

E9 is therefore a consistent direction rather than a demonstrated gain of a stated size, and we report it that way. A narrower margin is available from the higher sampling budget, 0.0236, and E9 clears that one, but the bar was set at 0.0315 before anything ran, and moving a bar after seeing the numbers would cost the sweep the only thing that makes it readable.

The one improvement we found does not survive a change of script
shaded band is the margin of error



5.3 E9 on English

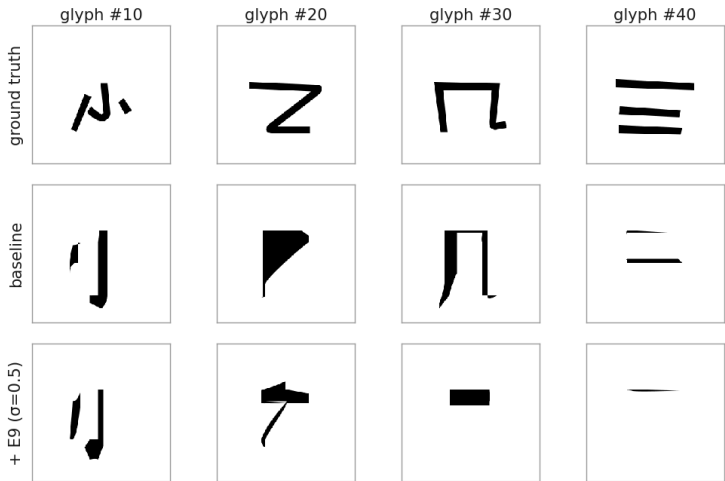
We repeated E9 on English at three seeds, paired the same way. One seed favours it on both metrics and two go against it, and every difference is inside the English margin of error.

This was written down as a possible outcome before the runs started, so it is a result rather than a failed experiment: **the one improvement we found does not survive a change of script.**

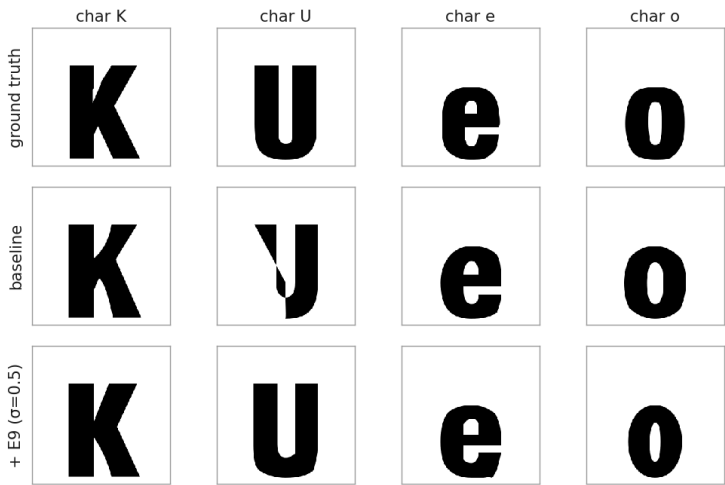
5.4 What the two models draw

Baseline and E9 side by side, ground truth alongside, same font and same four characters on both languages:

Chinese · font 0000 · mean L1 over these 6 fonts -- baseline 0.133, +E9 0.131



English · font 0000 · mean L1 over these 6 fonts -- baseline 0.064, +E9 0.066



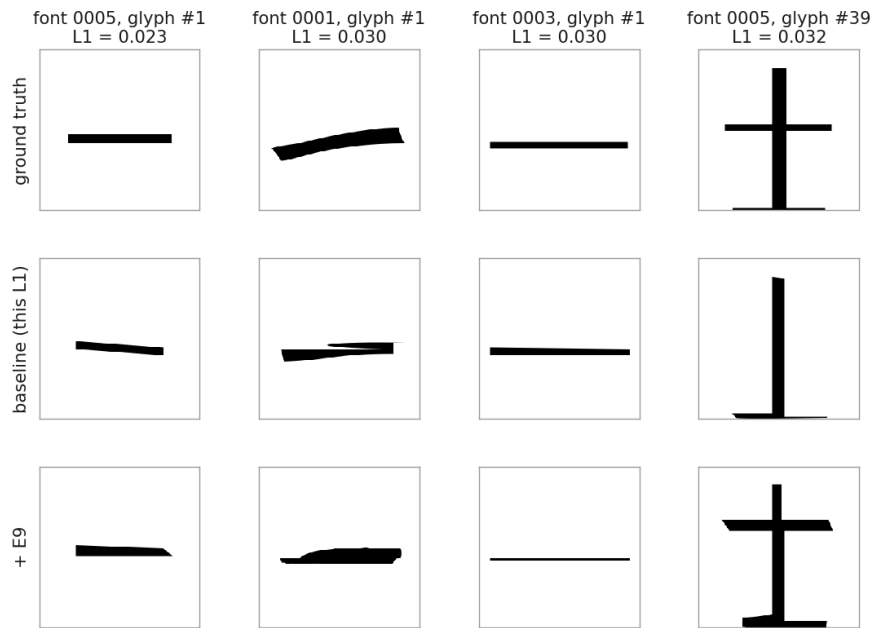
Same font, same four characters, both languages. Not a claim: section 5.2's reading is that E9's Chinese mean does not beat the margin of error, and a picture cannot upgrade that -- this is what the numbers in section 5 look like as glyphs, nothing more.

English is close to solved at this scale; Chinese is not, and the two E9 columns are close enough to each other that a reader should not expect to see the improvement in section 5.2 by eye: 0.0040 L1 is about 16 of the image's 4096 pixels, a difference the metric can pick up averaged over hundreds of glyphs but not one a single rendered comparison reliably shows. (The L1 figures above these panels are computed over the six fonts decoded for this figure alone, and are not meant to match the 34-font numbers in sections 3 and 5.1.)

The same six decoded Chinese fonts also bound the distribution behind that 0.163 mean.

Section 3 said an average hides the tail; here is both ends of it, picked by L1 and not by eye, same models as above. First the low end:

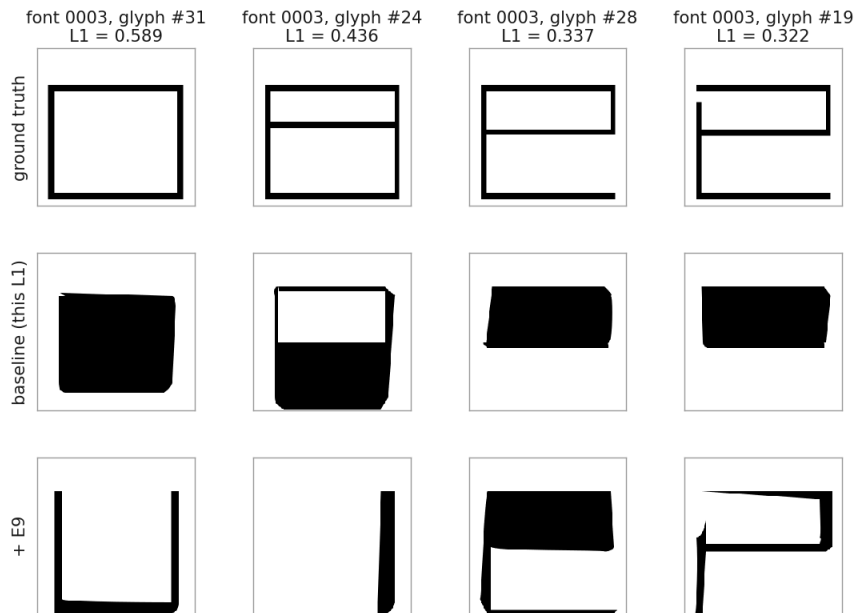
Best Chinese baseline glyphs by L1, out of the 6 decoded test fonts (chosen by the number, not by eye)



The other tail of the same distribution as the failure strip below: these are what the low end of a 0.163 Chinese mean looks like, picked by the identical rule -- lowest L1 among the same six decoded fonts, not the most flattering glyphs found by scanning. Simple strokes with little overlap area reconstruct close to the ground truth; the failure strip shows what the model does with denser, more overlapping strokes instead.

Then the high end:

Worst Chinese baseline glyphs by L1, out of the 6 decoded test fonts (chosen by the number, not by eye)



The reported Chinese Error is 0.163 averaged over 34 fonts; these are what 0.16-ish actually looks like at the tail, not the average case.

The pattern at the bad end is consistent: three of these four glyphs, all simple rectangular strokes in the ground truth, collapse into a solid filled block rather than an outline. E9 does

not repeat that pattern on the same four glyphs — it fills one, leaves one nearly blank, and draws thin partial outlines on the other two — a different failure, not a smaller version of the same one. At the good end the pattern is close to the mirror image: what survives is glyphs with little stroke overlap to begin with, single bars and simple crosses, which is a property of the glyph rather than something either model did well. Neither tail is an artifact of the rendering, since every comparison above renders the *same* outline on both sides, so what the panels show is the model drawing the right or the wrong thing. The spread between the two ends is the kind of difference a 64×64 pixel-disagreement count is well suited to catching.

5.5 Remaining candidates

Everything else was null. Below are the one-per-category representatives, three seeds each on Chinese, at matched checkpoint epoch, against a margin of error of 0.0097 on Error and 0.0315 on s-IoU.

Change	Mean Δ Error	Mean Δ s-IoU	Reading
Batch norm in the image branch (E2)	−0.0066	+0.0221	same sign, inside the margin
Encoder width (E4)	+0.0059	+0.0173	mixed sign
Latent 256 (E5)	−0.0010	+0.0203	mixed sign
Bézier weight 0.1 (E7)	−0.0010	−0.0053	mixed sign
AdamW (E11)	−0.0011	+0.0017	mixed sign
Refinement depth 2 (E3)	−0.0004	+0.0007	mixed sign
Encoder depth 8 (E16)	+0.0025	−0.0045	mixed sign
Decoder width 2048 (E17)	+0.0005	+0.0051	mixed sign
Terminal LayerNorm on the encoder (E1)	−0.0015	−0.0376	degrades, past the margin

E2 is the only null with a consistent sign on both metrics, and both of its means sit inside the margin, so it does not pass either. E1 is the one row that beats its margin, in the degrading direction, and the only effect anywhere in the project that does so.

Running the same changes on English disagreed with Chinese. Refinement depth 2 (E3) is mixed-sign and null on Chinese, and on English it beats the margin of error in the degrading direction at all three seeds. Latent 256 (E5) does the same. The same one-factor edit points in opposite directions on the two scripts the paper reports.

6. Discussion

6.1 Findings

The reproduction worked, and the way we checked it is the part worth keeping. Scoring the authors’ released weights through our own code turned “we fell short of 0.080” into a question with an answer: our Chinese training lands 0.0008 from their released checkpoint on a quarter of the budget, and our English training scores better than all three of theirs. Without a released checkpoint to score, the strongest conclusion available would have been “probably undertrained”, which is what earlier drafts of this work carried.

Measuring the seed spread before running any candidate cost three training runs on the first day and changed every conclusion after it.

Of the three candidates that came from reading the paper against its implementation, two produced our two clearest results. Reading a paper against its code is a better source of hypotheses than guessing at hyperparameters.

6.2 Null results

Eighteen of twenty-one changes did nothing measurable. We do not read this as evidence that the changes were badly chosen. We read it as evidence that the benchmark cannot resolve changes of this size.

Set our margins against the paper’s own tables. Its English ablation spans 0.0069 across four rows, with individual steps of 0.0031, 0.0028 and 0.0010, and its sampling-point study spans 0.0037 across six settings, with steps between 0.0002 and 0.0014. Our English margin of error is 0.0038 and our Chinese margin is 0.0097. **Most rows in the paper’s own ablations differ by less than our measured margin of error**, on the same metric and the same data.

That is not a claim that those results are wrong. It is a claim about what a single training run per configuration can support. Neither the paper nor the work it compares against reports a seed spread, so the published tables do not say either way.

E9 failing to transfer to English is the other thing that did not work, and it is informative rather than merely disappointing. The same holds for the direction disagreement in section 5.5, where one-factor edits point opposite ways on the two scripts the paper reports. A single language licenses less than a two-language paper makes it look like it does.

6.3 Methodological implications

A metric is a pipeline, not a formula. “Mean absolute error between two 64×64 images” is not enough information to reproduce a number. How many candidate outlines the best-of-N selection draws from is worth 0.0027 on English, a quantity that is pure sampling budget with no model in it, and larger than several of the differences the sweep set out to detect.

A single-seed baseline reports the draw, not the change. The 26 candidate configurations span 0.0101 between them; three re-runs of the unmodified model span 0.0097. Measured against a single baseline run, 22 of 26 candidates beat the baseline; measured against the mean of three, 6 of 26 do. The only difference between the two readings is that the single run happened to land 0.0048 above the mean, an ordinary draw. Ranking candidates on that single run and re-running the top three at two more seeds, one of three survived. This is the result we would most want someone else to check, and checking it is cheap. Take a published vector font ablation, retrain two rows at three seeds, and see whether the order holds.

6.4 Limitations

Every margin here rests on three seeds, so the bars themselves are uncertain. English was scored on 34 fonts rather than the full 1,386, and at 50 samples per glyph rather than the paper’s 10. Our Chinese training set is augmented six times where the paper says ten, as section 1.1 records. Nothing was run at more than three seeds.

6.5 Future work

More seeds first. Every bar in this report rests on three of them, which pins the margin down loosely. Chinese runs cost about an hour each, so five or ten seeds is an affordable way to tighten every comparison in section 5.

Then the output head. The decoder is autoregressive over quantized coordinate bins, and we measured what the quantization itself costs by encoding ground truth outlines into the model’s own representation and decoding them back with no model involved. The released 128-bin grid costs 0.0021 in L1 against unlimited precision, so the grid is not the bottleneck, but the sequential factorization over it may be. A flow-matching head over continuous coordinates would remove the quantization and the sequential decode together. We designed and costed one and dropped it on schedule grounds rather than on merit; it is in `archive/FLOW_MATCHING_PLAN.md`.

Beyond this model, the measurement itself is the transferable part. Any ablation table in this literature can be re-read the way section 6.3 re-reads ours, and the cost of doing so is two retrainings per row.

7. References

1. Wang, Y., Wang, Y., Yu, L., Zhu, Y., Lian, Z. *DeepVecFont-v2: Exploiting Transformers to Synthesize Vector Fonts with Higher Quality*. CVPR 2023. arXiv:2303.14585.
2. Wang, Y., Lian, Z. *DeepVecFont: Synthesizing High-Quality Vector Fonts via Dual-Modality Learning*. ACM TOG 2021.
3. Carlier, A., Danelljan, M., Alahi, A., Timofte, R. *DeepSVG: A Hierarchical Generative Network for Vector Graphics Animation*. NeurIPS 2020.
4. Lopes, R. G., Ha, D., Eck, D., Shlens, J. *A Learned Representation for Scalable Vector Graphics*. ICCV 2019.
5. Liu, Y., Guo, F., Wang, Z., Zhang, D. *DualVector: Unsupervised Vector Font Synthesis with Dual-Part Representation*. CVPR 2023. Source of the SSIM / L1 / s-IoU reporting convention.
6. Wang, Z., Bovik, A. C., Sheikh, H. R., Simoncelli, E. P. *Image Quality Assessment: From Error Visibility to Structural Similarity*. IEEE TIP 2004. SSIM.
7. Xiong, R., et al. *On Layer Normalization in the Transformer Architecture*. ICML 2020. For E1.
8. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z. *Rethinking the Inception Architecture for Computer Vision*. CVPR 2016. Label smoothing, for E8.
9. Loshchilov, I., Hutter, F. *Decoupled Weight Decay Regularization*. ICLR 2019. AdamW, for E11.
10. Higgins, I., et al. *β -VAE: Learning Basic Visual Concepts with a Constrained Variational Framework*. ICLR 2017. For E12.
11. Jaegle, A., et al. *Perceiver: General Perception with Iterative Attention*. ICML 2021. The architecture the encoder’s configuration advertises and does not instantiate.
12. Wilcoxon, F. *Individual Comparisons by Ranking Methods*. Biometrics Bulletin 1945. The paired test used per seed.
13. Lipman, Y., et al. *Flow Matching Guide and Code*. arXiv:2412.06264. On the course reading list; basis for the head proposed in section 6.5.

Code, data and checkpoints. Everything is at github.com/bensapirstein/deepvecfont-v2, a fork of github.com/yizhiwang96/deepvecfont-v2. Branch main is a pristine mirror of upstream and was never committed to, so **git diff main..submission is exactly what this project contributed**: 69 files, against the roughly 4,700 lines of released Python it sits on. The delivered tree is branch submission, tagged v1.0-submission. docs/REPRODUCE.md runs from a clean conda environment to a rebuilt version of this PDF.

The model code is one code path. The reconstruction and the improved model are the same train.py, models/ and test_few_shot.py; E9 is --enc_noise_std_train 0.5 on the training command and nothing else. That holds for all 21 candidates: each is a flag that defaults to the released behaviour, and a preflight script asserts every default reproduces the baseline exactly. It is a weaker guarantee than it sounds, since it cannot catch a change that is correct and irrelevant, but it does rule out the failure where a candidate appears to help because it quietly moved something else as well.

Data. The authors’ released Chinese and English sets, at their own download links, built by data_utils/, with every split committed as JSON in data_splits/ so that a number reported here is reported on a split anyone can reconstruct.

Checkpoints are on Google Drive, linked from the repository README: the Chinese baseline and Chinese E9, the English baseline and English E9, seed 1111 in each case, with the run’s checkpoint_metrics.csv alongside so the selection is auditable rather than asserted. The three-seed means in section 5.1 come from three such runs each; the released seed is the representative draw, and every scored checkpoint in the project has a row in RESULTS.csv.

What the repository deliberately does not carry. No decode trees, no weights in git, and none of the thirty days of runbooks, dated status tables and cluster notes the work actually generated. Those stay on the repro branch, which is public for anyone who wants the process rather than the result; docs/PROVENANCE.md on submission says where each citation in the code points. Model outputs regenerate from the commands, and what was measured from them is in RESULTS.csv instead.

Nothing in this report is typed twice. RESULTS.csv holds all 140 scored checkpoints, one row each. Every figure is regenerated from it by report/make_figures.py and report/make_model_comparison_figures.py, and every number quoted here is re-derived from it and checked by report/verify_report.py, 64 assertions that exit non-zero on any mismatch and that gate the PDF build. A figure or a number that has drifted from the results table stops the build rather than reaching a submission.